

PUOCH MABOR MAKUEI

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GitHub: github.com/puoch1of1 | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data-focused Computer Science finalist specialising in Data Science and Data Engineering, with hands-on experience in analytics, IT systems, and data quality assurance in both national and international contexts. Skilled in Python, SQL, and data visualisation, with growing expertise in geospatial analysis and data-driven decision support. Passionate about applying data to address climate, security, and development challenges across Africa. Demonstrated ability to work with international teams, manage data quality standards, and translate complex data into actionable insights. Seeking to leverage technical skills and regional knowledge to support UNMISS in climatesensitive conflict analysis and integrated data analytics.

EDUCATION

Uganda Christian University — Bachelor of Science in Computer Science

2023 – July 3rd, 2026 (Graduation)

Specialization: Data Science & Data Engineering

Relevant Coursework: Data Science, Data Analysis, Machine Learning, Statistics, Networking, Database Design and Programming

PROFESSIONAL EXPERIENCE

Statistics Without Borders (SWB) — Deliverable Quality Assurance Volunteer

January 2025 – Present (Remote)

- Validate datasets, analytical reports, and deliverables to ensure accuracy, consistency, and adherence to international quality standards in humanitarian and development contexts
- Collaborate with geographically dispersed international teams to assess data quality, identify inconsistencies, and recommend improvements across structured and semi-structured data sources
- Apply analytical thinking to document quality assurance workflows and support data-driven decision-making processes

National Communication Authority (NCA) — IT Support & Systems Intern

May 2025 – September 2025 | Juba, South Sudan

- Analyzed network performance data and diagnosed VLAN, Ethernet, and firewall issues; resolved connectivity problems affecting multiple departments
- Operated spectrum monitoring tools to collect, interpret, and analyze signal data; identified interference patterns and field anomalies for operational decision-support
- Contributed to enterprise network architecture design using subnetting (VLSM), VLANs, and access control lists for secure service delivery
- Applied cybersecurity analysis techniques using Kali Linux and Wireshark to assess network vulnerabilities and support threat assessment workflows

PROJECTS & TECHNICAL WORK

Comprehensive Data Analytics & Engineering Portfolio | github.com/puoch1of1

A portfolio of data analytics, engineering, and applied data science projects focused on ETL pipelines, data cleaning, API ingestion, database management, exploratory analysis, dashboarding, and time series forecasting using Python and SQL.

World Bank API ETL Pipeline — Data Extraction & SQL Loading

- Developed production-grade ETL pipeline to extract South Sudan socioeconomic indicators (GDP, population, life expectancy) from REST API (1960–2024)
- Transformed flat API responses into normalized, tidy records; loaded into SQLite with data quality validation and verification checks
- Demonstrated end-to-end data pipeline competency: ingestion, transformation, validation, and persistence

Geospatial Analysis: Climate & Conflict Trends in South Sudan (In Progress)

- Conducting spatial analysis using QGIS to examine relationships between rainfall variability and conflict incident patterns using ACLED and climate datasets
- Producing layered geospatial visualizations to identify regional patterns, seasonal trends, and hotspots for situational awareness
- Integrating multi-source data (precipitation, vegetation, conflict) to explore climate-security nexus and inform risk assessment

TECHNICAL SKILLS

Geospatial & Spatial Analysis: QGIS (vector and raster analysis), spatial data handling, geographic data visualization, geospatial data processing

Programming & Database: Python (Pandas, NumPy, Scikit-learn), SQL (MySQL, PostgreSQL), JavaScript

Data Visualization & Analytics: Matplotlib, Seaborn, data visualization design, exploratory data analysis

Data Management: Data quality assurance, data documentation, structured and semi-structured data, data cleaning workflows

Domain Knowledge: Statistics, Machine Learning fundamentals, humanitarian data contexts, climate-security analysis

Tools & Platforms: Git/GitHub, Jupyter Notebook, VS Code, MySQL/PostgreSQL, Microsoft Office

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Machine Learning in Python (IBM Coursera, 2025)
- Commonwealth Bank Cybersecurity Simulation (Forage, 2025)
- CS50 – Introduction to Computer Science (edX, 2024)
- Programming Essentials in Python (Cisco, 2024)

LANGUAGES

English (Fluent) • **Dinka** (Intermediate)

KEY STRENGTHS

- Regional expertise in South Sudan and East Africa; familiar with humanitarian and development sector dynamics
- Experience working across international teams in remote and in-country contexts
- Strong foundation in data quality, documentation, and analytical workflows aligned with UN standards.